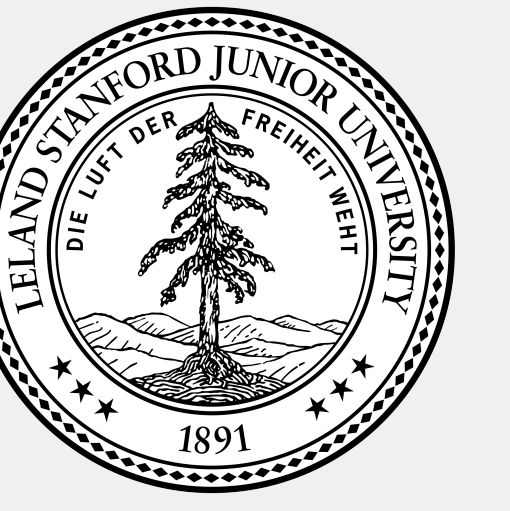




Bayesian Learning in Online Decision Making

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Value of Offline vs Online Learning for Decision-Making

Online decision-making under uncertainty raises a fundamental question about the relationship between *learning* and *decision-making*: when an algorithm begins interacting with an environment, should it

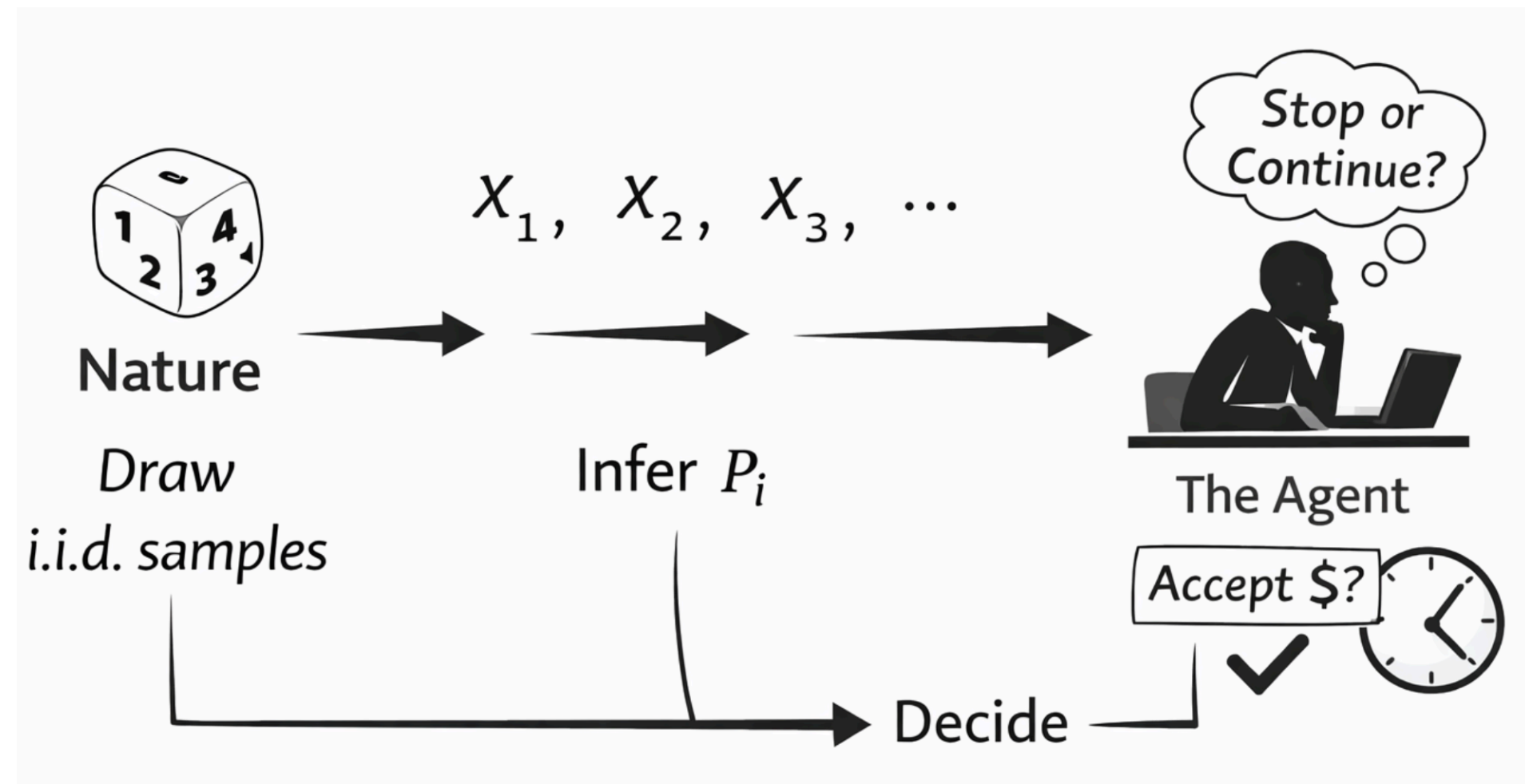
- rely on **prior knowledge** (training data),
- rely on **online samples**,
- or combine both in a **Bayesian** way?

While *be Bayesian* is often good advice, the value of prior knowledge versus online learning could vary dramatically, and in some settings, one of these sources of information is redundant for making the optimal decision.

Classical Models for *Beyond Worst-Case* Algorithms

Classical models for online algorithms under uncertainty sit at two opposite extremes and **often obscure the value of online learning**. (This tension is especially sharp in optimal stopping, in which the decision-maker observes a sequence of n i.i.d. samples X_t drawn from a prior distribution online and want to stop on the largest possible observation.)

- Assuming a known distribution (**prophet inequality**) is overly optimistic and, under this model, there is nothing to learn.
- Adversarial models tend to be too pessimistic: For example, [2, Theorem 2]. showed that, for the optimal stopping problem under an unknown i.i.d. distribution, learning from early samples alone cannot improve on the benchmark of $1/e$ for when samples are exchangeable but completely adversarial otherwise (**secretary problem**).



Bayesian Model

We introduce a Bayesian perspective to this landscape that interpolates smoothly between the classical known distributional model and models with no known or many competing possible distributions!

There are k known possible distributions to the decision-maker

$\mathcal{P}_1, \dots, \mathcal{P}_k$, and a known, correct prior $\pi \in \Delta_k$

over these candidates. Nature draws a distribution $I \sim \pi$, and conditioned on I , the decision-maker observes

$$X_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{P}_I, \quad t = 1, \dots, n.$$

single prediction
known i.i.d.
prophet inequalities
consistency
CR: 0.745

$k = 3$

infinite/no predictions
unknown i.i.d.
secretary
robustness
CR: $1/e$

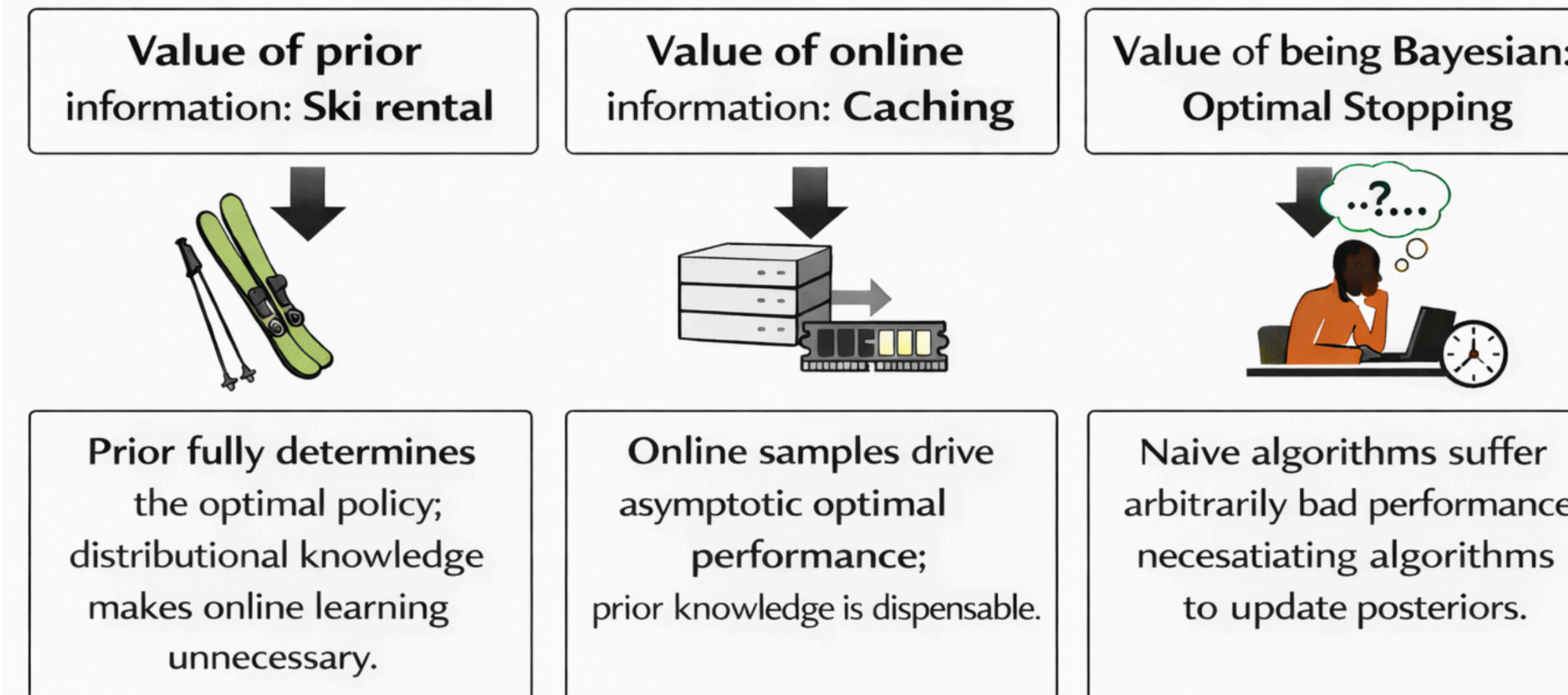
Performance Metric: Competitive Ratio

For a fixed online algorithm ALG, we define $\mathbb{E}_\pi[\text{ALG}]$ to be its expected value under π . Then, the competitive ratio of an algorithm is defined as

$$\text{CR}_\pi(\text{ALG}) := \frac{\mathbb{E}_\pi[\text{ALG}]}{\mathbb{E}_\pi[\text{OFF}]},$$

where the optimum offline OFF is the value of the best decision in hindsight.

Different Regimes



- Ski rental (prior only):** Let $T \in [n]$ be the random horizon with an arbitrary prior distribution and $\mathcal{H}_t$ be the information available at time t . Then, for every set $A \subseteq [n]$,
$$\Pr(T \in A \mid \mathcal{H}_t, T \geq t) = \Pr(T \in A \mid T \geq t).$$
- Caching (samples only):** Under i.i.d. requests, a simple algorithm (using heavy-hitters methods) that behaves like the well-known LFU (Least Frequently Used) policy based only on online samples can drive asymptotically optimal performance.

Bayes-Optimal Dynamic Program $\rightarrow$ Approximation Algorithms

Define the *value-to-go* from state $(t, \mathcal{D}(t))$ recursively as

$$J_t(\mathcal{D}(t)) = \mathbb{E}_{X_{t+1} \sim \mathcal{P}_{\mathcal{D}(t)}}[\max\{X_{t+1}, J_{t+1}(\mathcal{D}(t+1) \mid X_{t+1})\}],$$

with $J_{n+1}(\mathcal{D}) = 0$. Here, $\mathcal{D}(t+1) \mid X_{t+1}$ denotes the posterior. The optimal online policy is therefore: at time t , given posterior $\mathcal{D}_{\text{belief}}(t)$, stop if and only if

$$X_t \geq J_t(\mathcal{D}_{\text{belief}}(t)).$$

Importantly, the optimal policy jointly depends on the randomness of future observations and the induced stochastic evolution of posterior beliefs, which is **computationally intractable** to capture.

Research Question

This raises a **computational question**: must one compute the fully Bayesian optimal policy—often intractable (indeed, PSPACE-hard in general [4, Theorem 6])—or can *approximately Bayesian* algorithms achieve near-optimal performance efficiently?

Bayesian online is as *hard* as unknown online

By Yao's minimax principle,

$$\min_{\pi} \max_{\text{ALG}} \text{CR}_\pi(\text{ALG}) = \max_{\text{ALG}} \min_{\pi} \text{CR}_\pi(\text{ALG}).$$

- While this implies that we cannot beat the worst-case bounds given for when the underlying distribution is unknown, it does not eliminate the possibility for an improvement in our Bayesian model when the support size k is finite.

Existing policies perform poorly $\rightarrow$ need for adaptivity

Ex: Set the threshold at time t as the $(1 - \frac{1}{n})$ -th quantile of the posterior predictive at that time (this policy gives optimal CR in prophet setting!) but...

Distribution \ Values	$S = 0$	$M = 1$	$L = n^5$
$\mathcal{P}_1$	$1 - \frac{1}{n}$	$\frac{1}{n}$	0
$\mathcal{P}_2$	$1 - \delta - \frac{1}{n}$	δ	$\frac{1}{n}$

Table: Probability Mass Functions for Distributions $\mathcal{P}_1$ and $\mathcal{P}_2$ over $S < M < L$

Proposition (Static threshold policies with random tie-breaking). For any $\varepsilon > 0$, threshold τ , and $p \in [0, 1]$, there exists a prior distribution such that the following policy has competitive ratio at most $O(\varepsilon)$ against the offline optimum: always accept if $X_t > \tau$, accept X_t with probability p if $X_t = \tau$, and reject otherwise.

- This implies that our setting is harder.

Polynomial Time Approximation Scheme

Assuming $\{X_i\}_{i=1}^n \subseteq [u, U]$, for any target accuracy $\varepsilon \in (0, 1)$, there is an online policy computable by a dynamic program with

$$O\left(\left(\frac{Cn^2}{\varepsilon} \log\left(\frac{nUk}{\varepsilon u}\right)\right)^k\right)$$

states (and hence the same time complexity), for some absolute constant C , whose value-to-go at every time t is a $(1 \pm \varepsilon)$ multiplicative approximation of the value-to-go of the Bayes-optimal stopping algorithm.

- Key idea:** *Discretize/round* posterior probabilities onto a finite grid + run a finite-state DP over the discretized belief space to obtain value-to-go that is within a $(1 \pm \varepsilon)$ multiplicative factor.

Algorithm $\rightarrow$ lower bound

For every fixed $k \geq 1$ and every instance $(\pi, \{\mathcal{P}_h\}_{h=1}^k)$, there exist a polynomial time algorithm ALG such that

$$\text{CR}_\pi(\text{ALG}) \geq \frac{1}{e} + \Omega\left(\frac{1}{k^2}\right).$$

- Key idea:** Doubling trick over the expected contribution of each distribution to OFF + careful setting of consistency & robustness parameters in [1, Theorem 1].
- Asymptotically tight** as $k \rightarrow \infty$ by [3, Theorem 5.1].
- Implies that the existential $1/e$ bound given in [3] is only true at $k = \infty$.
- Resolves an **open problem** from [3].

Hardness $\rightarrow$ upper bound

For every $k \geq 1$, there exists an instance $(\pi, \{\mathcal{P}_h\}_{h=1}^k)$ and $N_0 \in \mathbb{N}$ such, that for all $n \geq N_0$, any online algorithm ALG satisfies

$$\text{CR}_\pi(\text{ALG}) \leq \frac{1}{e} + O\left(\frac{n \log(k) + n \log n}{\sqrt{k}}\right).$$

- Gives **constructive proof** for [3, Theorem 5.1] on the way.

Prior	Dist \ Val	1	n^3	n^6	n^9	$\dots$	$n^{3(k-1)}$	n^{3k}
$1 - \sum_{i=1}^{k-1} \frac{1}{n^{3i}}$	$\mathcal{P}_1$	$1 - \frac{1}{n^2}$	$\frac{1}{n^2}$					
$\frac{1}{n^3}$	$\mathcal{P}_2$	$(1 - \frac{1}{n^2})^2$	$(1 - \frac{1}{n^2}) \frac{1}{n^2}$	$\frac{1}{n^2}$				
$\frac{1}{n^6}$	$\mathcal{P}_3$	$(1 - \frac{1}{n^2})^3$	$(1 - \frac{1}{n^2})^2 \frac{1}{n^2}$	$(1 - \frac{1}{n^2}) \frac{1}{n^2}$	$\frac{1}{n^2}$			
$\vdots$	$\vdots$							
$\frac{1}{n^{3(k-2)}}$	$\mathcal{P}_{k-1}$	$(1 - \frac{1}{n^2})^{k-1}$	$(1 - \frac{1}{n^2})^{k-2} \frac{1}{n^2}$	$(1 - \frac{1}{n^2})^{k-3} \frac{1}{n^2}$	$\dots$	$(1 - \frac{1}{n^2}) \frac{1}{n^2}$	$\frac{1}{n^2}$	
$\frac{1}{n^{3(k-1)}}$	$\mathcal{P}_k$	$(1 - \frac{1}{n^2})^k$	$(1 - \frac{1}{n^2})^{k-1} \frac{1}{n^2}$	$(1 - \frac{1}{n^2})^{k-2} \frac{1}{n^2}$	$\dots$	$\dots$	$(1 - \frac{1}{n^2}) \frac{1}{n^2}$	$\frac{1}{n^2}$

Table: Probability Mass Functions for Distributions

Next steps: Online Steiner Tree

- Steiner tree involves many incremental decisions as opposed to a single irrevocable decision in optimal stopping.*
- Posterior convergence** actually makes a difference here!
- Myopic** policies that only use the current posterior $\mathcal{D}_{\text{belief}}(t)$ but ignore the fact that continuing at time t endogenously changes future beliefs $\mathcal{D}_{\text{belief}}(t+1), \mathcal{D}_{\text{belief}}(t+2), \dots$ could work!

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